Easy Visa

Advanced Machine Learning

3 May 2026

Contents / Agenda

Executive Summary

- The growing demand for skilled labor in the United States has led to a significant increase in visa applications processed by the Office of Foreign Labor Certification (OFLC)
- Manual evaluation of these applications is becoming time-consuming and inefficient, creating a need for a scalable, data-driven solution
- This project uses machine learning to predict visa certification outcomes and identify the key factors influencing approval decisions
- Analysis of 25,000+ applications highlights the importance of factors such as education, work experience, job type, and wage characteristics
- The final model enables efficient shortlisting of high-probability candidates, improving decision-making consistency
- Implementing this solution can help:
    - Reduce processing time
    - Improve accuracy and consistency in approvals
    - Support better workforce planning for employers

Business Problem Overview

Business Problem Overview:

- U.S. business communities face increasing demand for skilled labor and often rely on foreign workers to address workforce shortages
- Under the Immigration and Nationality Act (INA), employers must demonstrate that hiring foreign workers will not negatively impact U.S. workers
- The Office of Foreign Labor Certification (OFLC) is responsible for reviewing and certifying visa applications

Challenge:

- In FY 2016 alone:
    - 775,979 applications processed
    - 1.69 million positions requested
- The volume of applications is increasing each year, making manual review inefficient and resource-intensive

Key Problem:

- Difficulty in efficiently identifying candidates with a high likelihood of visa approval

Solution Approach and Outcome

Solution Approach

- Develop a machine learning classification model to predict visa certification outcomes
- Perform Exploratory Data Analysis (EDA) to identify patterns and key drivers of approval
- Apply data preprocessing techniques to prepare data for modeling
- Build multiple classification models, including:
    - Decision Tree
    - Random Forest
    - Bagging and Boosting methods
- Address class imbalance using:
    - Oversampling and under sampling techniques
- Optimize model performance using:
    - Cross-validation
    - Hyperparameter tuning

Outcome:

- Provide a data-driven system to:
    - Facilitate faster visa approvals
    - Recommend suitable applicant profiles
    - Support consistent and scalable decision-making

Data Overview

Overview of the Dataset
The dataset consists of **25,480 visa application records** with **12 variables** capturing employee, employer, and job-related information.

Preview of the Data
A snapshot of the dataset (first and last 5 rows) shows the structure and types of information available. Each row represents a unique visa application and includes:
- **Employee details**: continent, education level, work experience, training requirement
- **Employer details**: company size (number of employees), year established
- **Job details**: region of employment, wage, wage unit, full-time status
- **Target variable**: case_status (Certified or Denied)

From the sample data:
- Applicants come from multiple continents, with a strong presence from Asia
- Education levels vary from High School to Doctorate
- Wage values vary significantly depending on unit (hourly vs yearly)
- Both certified and denied cases are present in the dataset

Dataset Dimensions
- **Number of observations (rows):** 25,480
- **Number of features (columns):** 12

Key Observations
- Each record is uniquely identified by a case_id
- The dataset contains a mix of categorical and numerical variables
- The target variable case_status indicates whether a visa was approved (Certified) or rejected (Denied)
- There is noticeable variation in wages, company sizes, and employment regions, which may influence visa outcomes

Summary
Overall, the dataset provides a comprehensive view of visa applications, combining demographic, economic, and employment-related factors. These variables will be used to identify patterns and build predictive models for visa approval.

EDA Results – Descriptive Statistics

```
[14]: data.describe() ##  Complete the code to print the st
```

[14]:

	no_of_employees	yr_of_estab	prevailing_wage
count	25480.000000	25480.000000	25480.000000
mean	5667.043210	1979.409929	74455.814592
std	22877.928848	42.366929	52815.942327
min	-26.000000	1800.000000	2.136700
25%	1022.000000	1976.000000	34015.480000
50%	2109.000000	1997.000000	70308.210000
75%	3504.000000	2005.000000	107735.512500
max	602069.000000	2016.000000	319210.270000

.

1. Number of Employees

•The average number of employees is ~5,667, while the median is much lower at ~2,109

•This indicates a right-skewed distribution, where a few very large companies significantly increase the average

•The maximum value (602,069) confirms the presence of large outliers

•Negative values were observed and corrected using absolute transformation

Insight: Most visa applications come from small to mid-sized companies, with a few large organizations contributing to extreme values.

2. Year of Establishment

•The average year of establishment is ~1979, while the median is ~1997

•The dataset includes companies ranging from very old (1800) to recent (2016)

Insight: Visa applications are submitted by a diverse mix of both well-established and relatively new companies

3. **Prevailing Wage**

- The average wage is ~$74,456, with a median of ~$70,308
- The wide range (from ~$2 to ~$319,210) indicates high variability in wages
- The distribution is slightly right-skewed, with some high-paying roles

Insight: Wage levels vary significantly across applications and may play an important role in determining visa approval outcomes.

EDA Results: Handling of Negative Employee Numbers

Handling Invalid Values in no_of_employees
I identified records where the number of employees was negative, which is not logically valid.
A total of **33 rows** contained negative values in this column.

Why this matters:
- Negative employee counts are data errors and can distort analysis and model performance

Action Taken:
- I applied an absolute value transformation to convert negative values into positive values

Rationale:
- I assumed that the magnitude of the value is correct and only the sign is incorrect

Outcome:
- Data quality is improved without losing any record

Fixing the negative values in number of employees columns

```
data.loc[data["no_of_employees"] < 0].shape ## Complete the code to check negative values in the employee column
```

```
(33, 12)
```

```
# taking the absolute values for number of employees
data["no_of_employees"] = abs(data["no_of_employees"])
```

EDA Results – Analysis of Categorical Data

Summary

This analysis highlights key patterns in the dataset, including dominance of certain categories and class imbalance. These insights will guide feature selection, preprocessing, and model development.

To better understand the dataset, I examined the distribution of all categorical variables by counting the frequency of each category.

Why this is important:

•Helps identify dominant patterns in the data

•Detects class imbalance

•Provides insights into factors that may influence visa approval

Key Observations

Case ID

•Each value appears only once (25,480 unique values)

•This confirms it is a unique identifier

Action:

•I removed this column as it does not contribute to prediction

Continent

•Majority of applicants are from Asia (~66%)

Insight:

•The dataset is geographically imbalanced, which may influence model predictions

Education Level

- Most applicants have Bachelor's or Master's degrees

Insight:

- The dataset is dominated by highly educated individuals

Job Experience

- More applicants have prior work experience than not

Insight:

- Work experience may be an important factor in visa approval

Job Training Requirement

•Most applicants do not require job training (~88%)

Insight:

•The majority of candidates are job-ready

Region of Employment

•Applications are distrib uted across major U.S. regions

Insight:

•Demand for foreign workers exists across multiple regions

Unit of Wage

•Majority of wages are yearly-based (~90%)

Insight:

•Most roles are stable, full-time salaried positions

Full-Time Position

•Most roles are full-time (~89%)

Insight:

•Visa applications are primarily for long-term employment

Case Status (Target Variable)

•Certified: 17,018

•Denied: 8,462

Insight:

•The dataset is moderately imbalanced, with more approvals than denials

•This imbalance will need to be addressed during model building

EDA Results – Univariate Analysis (Educational Level Distribution)

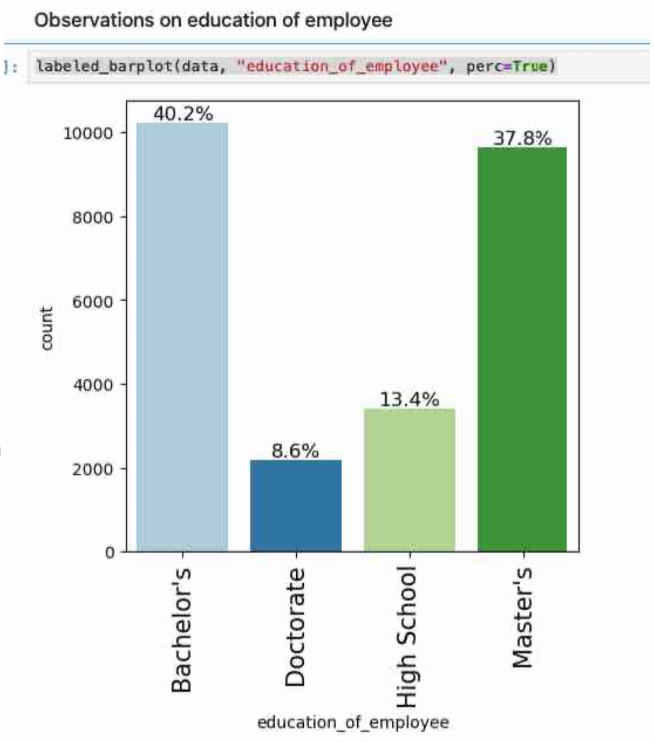
Education Level Distribution
The distribution of applicants by education level shows that:
- Bachelor's degree holders account for 40.2% of applicants
- Master's degree holders account for 37.8%
- High School graduates account for 13.4%
- Doctorate holders account for 8.6%

Interpretation
- The dataset is heavily concentrated around mid-level higher education (Bachelor's and Master's), which together make up nearly 78% of all applicants
- There is relatively low representation of applicants at the extremes (High School and Doctorate levels)

Why This is Important
- Education level is likely a key determinant of visa approval, as higher education is often associated with specialized skills and higher-paying jobs
- The imbalance in categories means the model will have more exposure to mid-level education groups, which may influence predictions

Business Insight
- The high proportion of Bachelor's and Master's degree holders suggests that visa applications are primarily driven by skilled professionals
- Employers are likely targeting candidates with strong academic backgrounds to meet workforce demands
- Understanding how education impacts approval can help:
    - Improve candidate screening
    - Align hiring with skill requirements
    - Increase approval success rates

Summary
Education level is a critical feature in the dataset, with most applicants being well-qualified. This reinforces the importance of education as a driver of visa approval outcomes.



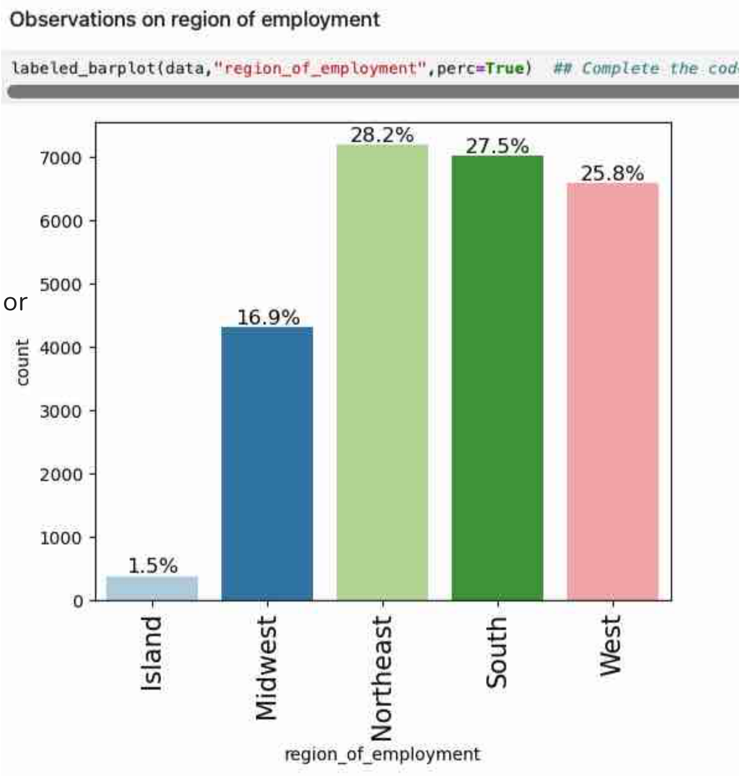
EDA Results – Univariate Analysis (Regional Employment)

Region of Employment Distribution¶
The distribution of visa applications across regions shows that:
- Northeast (28.2%), South (27.5%), and West (25.8%) account for the majority of applications
- Midwest (16.9%) has a moderate share
- Island regions (1.5%) have very few applications

Interpretation
- Visa applications are concentrated in major economic regions of the United States
- The Northeast, South, and West together account for over 80% of total applications, indicating higher demand for foreign workers in these areas
- The very low representation in Island regions suggests limited employment opportunities or lower demand in those areas

Why This is Important
- The region of employment may influence visa approval due to:
  - Differences in industry presence (e.g., tech, finance, manufacturing)
  - Regional labor shortages
  - Economic activity levels
- Since the data is not evenly distributed across regions, the model may learn stronger patterns from the dominant regions

Business Insight
- Employers in the Northeast, South, and West are the primary drivers of visa applications, likely due to higher economic activity and demand for skilled labor
- Understanding regional demand can help:
  - Target hiring strategies
  - Optimize workforce planning
  - Improve visa approval predictions based on location trends



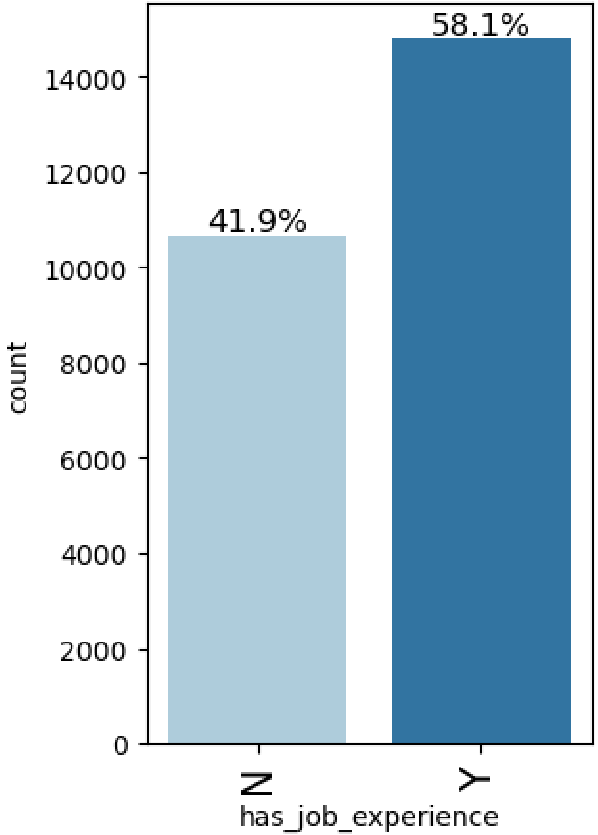
EDA Results – Univariate Analysis (Job Experience)

Job Experience Distribution
The distribution of job experience among applicants shows that:
- 58.1% of applicants have prior job experience (Y)
- 41.9% of applicants do not have prior job experience (N)

Interpretation
- A majority of applicants have relevant work experience, indicating that most candidates are already skilled professionals
- However, a substantial portion (~42%) lacks prior experience, suggesting a mix of entry-level and experienced applicants

Why This is Important
- Job experience is likely a key factor influencing visa approval, as employers may prefer candidates who can contribute immediately
- The presence of both experienced and inexperienced candidates allows the model to learn how experience impacts approval outcomes

Business Insight
- Employers appear to favor candidates with prior experience, which may increase the likelihood of visa certification
- Identifying the role of experience can help:
    - Improve candidate shortlisting
    - Align hiring strategies with workforce needs
    - Support more accurate visa approval predictions



EDA Results – Univariate Analysis (Case Status)

Case Status Distribution

The distribution of visa application outcomes shows that:

- 66.8% of applications are Certified
- 33.2% of applications are Denied

Interpretation

- A significantly larger proportion of visa applications are approved rather than denied
- This indicates that the dataset is moderately imbalanced, with approvals forming the majority class

Why This is Important

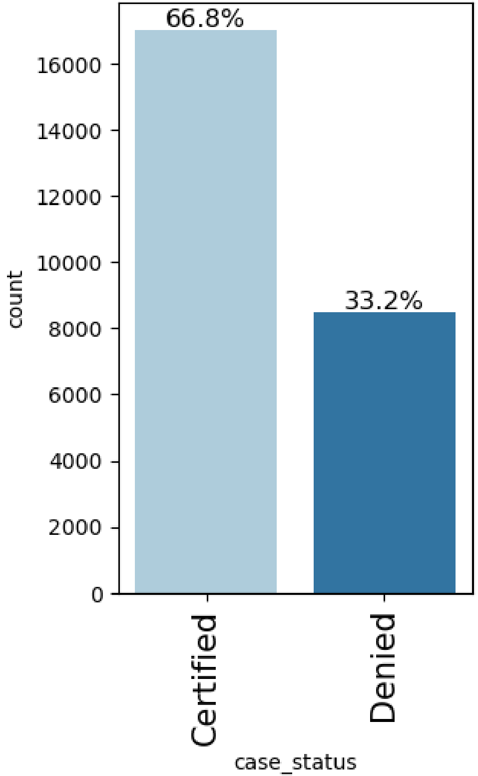
- Class imbalance can impact model performance:
  - The model may become biased toward predicting the majority class (Certified)
  - It may perform poorly in identifying denied cases
- This imbalance needs to be addressed during model building using techniques such as:
  - Oversampling
  - Under sampling

Business Insight

- The relatively high approval rate suggests that many applicants meet the required criteria for certification
- However, identifying the factors that lead to denial is equally important, as it helps:
  - Improve candidate screening
  - Reduce unsuccessful applications
  - Enhance decision-making efficiency

Summary

The imbalance in case status highlights the importance of building a model that can accurately predict both approvals and denials, ensuring balanced and reliable decision-making.



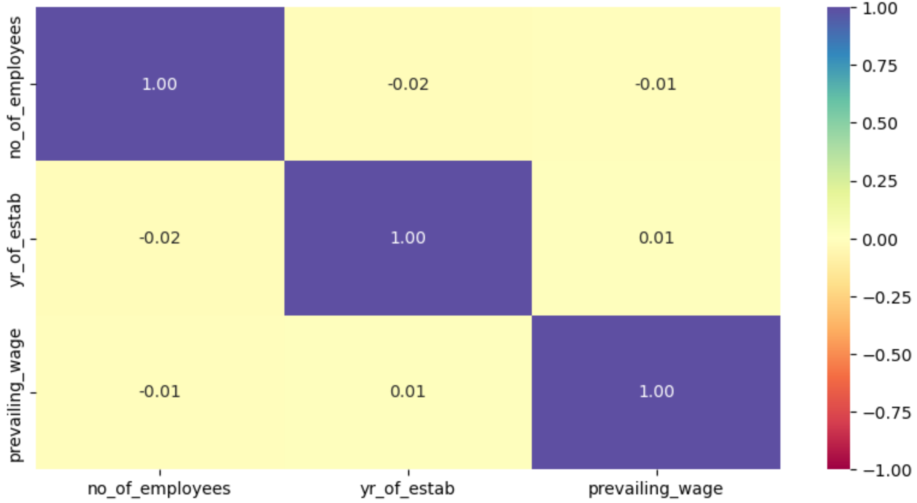
EDA Results – Bivariate Analysis (Correlation Check)

Correlation Analysis of Numerical Variables
A correlation heatmap was used to examine the relationships between numerical variables:
- no_of_employees
- yr_of_estab
- prevailing_wage

Key Observations
- The correlation values between variables are very close to **0**:
  - no_of_employees vs yr_of_estab: -0.02
  - no_of_employees vs prevailing_wage: -0.01
  - yr_of_estab vs prevailing_wage: 0.01

Interpretation
- There is **no strong linear relationship** between any pair of numerical variables
- The variables appear to be **independent of each other** in terms of linear correlation

Why This is Important
- Low correlation means:
  - No multicollinearity issues
  - Each variable contributes **unique information** to the model
- This is beneficial because:
  - Highly correlated variables can distort model performance
  - Independent features improve model reliability and interpretability



Business Insight
- Company size (no_of_employees), company age (yr_of_estab), and wage levels (prevailing_wage) operate independently
- This suggests that:
  - Large companies do not necessarily offer higher wages
  - Older companies are not strongly associated with higher or lower wages
- Each factor may influence visa approval **in different ways**, rather than being interconnected

Summary
The correlation analysis indicates that there are no strong relationships between numerical variables. This suggests that all features can be retained for modeling, as they provide distinct and non-redundant information

EDA Results: Bivariate Analysis -- Does education level influence visa approval?

Education Level vs Visa Approval (Case Status)

To understand whether education level influences visa approval, I analyzed the relationship between education_of_employee and case_status using a stacked bar plot.

Key Observations¶

From the distribution:

- Doctorate holders have the highest approval rate (~87%)
- Master’s degree holders also show a high approval rate (~79%)
- Bachelor’s degree holders have a moderate approval rate (~62%)
- High School applicants have a significantly lower approval rate (~34%)

Interpretation

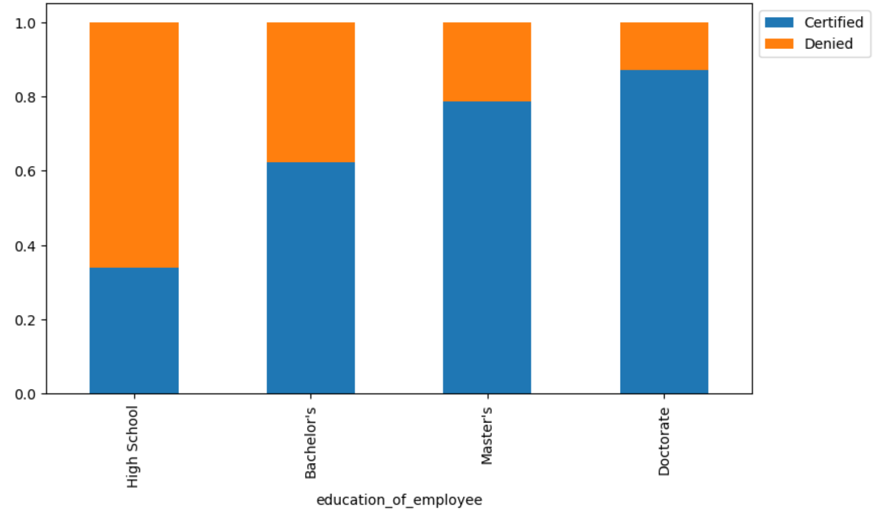
- There is a clear positive relationship between education level and visa approval
- As the level of education increases, the likelihood of visa certification also increases
- High School applicants are more likely to be denied compared to other groups

Why This is Important

- This confirms that education is a strong predictor of visa approval
- It highlights that employers and certification authorities likely favor:
  - Highly educated candidates
  - Individuals with specialized skills
- This variable will likely be a key feature in the model

Business Insight

- Employers seeking visa approval should prioritize candidates with:
  - Master’s or Doctorate degrees
- Candidates with lower education levels may face higher rejection rates and should:
  - Improve qualifications
  - Gain additional skills or certifications
- This insight can help:
  - Improve candidate screening
  - Increase visa approval success rates
  - Align hiring strategies with approval patterns




```
stacked_barplot(data, "education_of_employee", "case_status")

case_status            Certified  Denied    All
education_of_employee
All                        17018    8462  25480
Bachelor's                  6367    3867  10234
High School                 1164    2256   3420
Master's                    7575    2059   9634
Doctorate                   1912     280   2192
------------------------------------------------------------
```

Summary

Education level has a significant impact on visa approval outcomes, with higher education strongly associated with increased chances of certification. This makes it one of the most important factors in predicting visa approval.

EDA Results – Bivariate Analysis: How does visa status vary across different continents?

Continent vs Visa Approval (Case Status)
To understand how visa approval varies across different regions, I analyzed the relationship between continent and case_status.
Key Observations
- Europe has the highest approval rate (79%)
- Africa (72%) and Asia (65%) also show relatively strong approval rates
- North America (62%) and Oceania (64%) show moderate approval rates
- South America has the lowest approval rate (58%)

Interpretation
- Visa approval rates vary across continents, indicating that geographic origin influences certification outcomes
- Applicants from Europe and Africa have higher approval rates compared to other regions
- South America shows relatively lower approval rates, suggesting possible differences in applicant profiles or job types

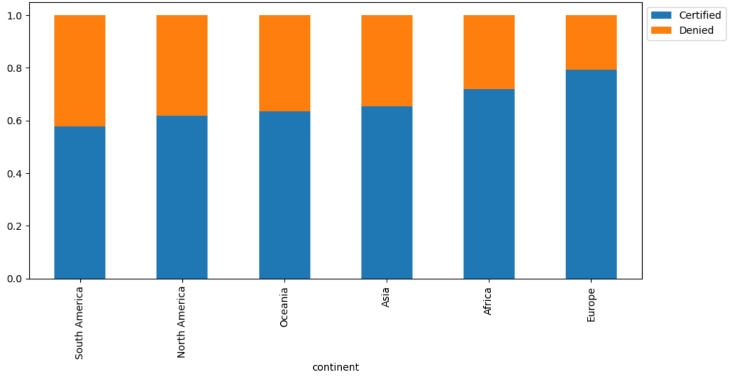
Why This is Important
- This shows that continent is an influential categorical variable in predicting visa approval
- Geographic differences may reflect:
    - Variation in skill levels
    - Industry demand
    - Types of roles applied for
- The model can leverage this information to improve prediction accuracy

Business Insight
- Employers may experience higher success rates when hiring candidates from regions with historically higher approval rates
- Regions like Europe and Asia appear to be strong sources of approved candidates, indicating alignment with U.S. labor market demands
- Lower approval regions may require:
    - Better candidate screening
    - Improved qualification matching

Summary
Visa approval rates differ across continents, with Europe and Africa showing higher certification rates and South America showing lower rates. This suggests that geographic origin plays a role in visa outcomes and should be considered in decision-making and predictive modeling.

```
stacked_barplot(data,"continent", "case_stat
```

case_status / continent	Certified	Denied	All
All	17018	8462	25480
Asia	11012	5849	16861
North America	2037	1255	3292
Europe	2957	775	3732
South America	493	359	852
Africa	397	154	551
Oceania	122	70	192



EDA Results – Bivariate Analysis: How does visa status vary across different continents?

Job Experience vs Visa Approval (Case Status)
To evaluate whether prior work experience influences visa approval, I analyzed the relationship between has_job_experience and case_status.

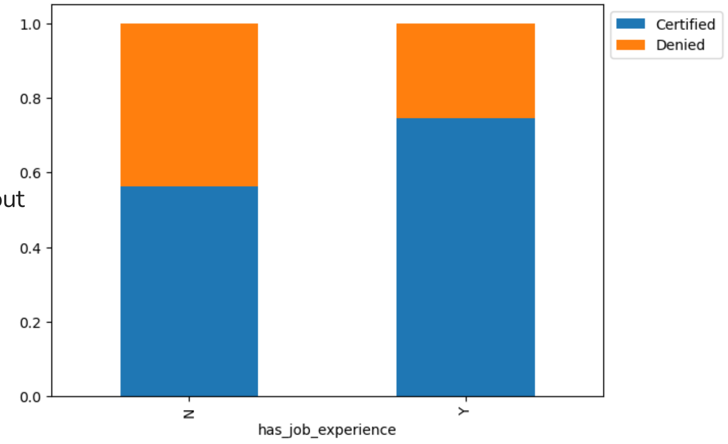
Key Observations
- Applicants **with prior work experience (Y)** have a higher approval rate (~74%)
- Applicants **without work experience (N)** have a lower approval rate (~56%)

Interpretation
- There is a clear positive relationship between **work experience and visa approval**
- Candidates with prior experience are significantly more likely to be certified compared to those without experience

Why This is Important
- Work experience is a strong predictor of visa approval
- Employers and certification authorities likely prefer candidates who can:
  - Contribute immediately
  - Require less training
  - Bring proven skills
- This variable will play an important role in model predictions

Business Insight
- Employers should prioritize candidates with prior work experience to improve approval success rates
- Candidates without experience may face higher rejection rates and may benefit from:
  - Gaining relevant experience
  - Improving skill sets
- This insight can help:
  - Improve candidate shortlisting
  - Reduce visa rejection rates
  - Align hiring strategies with approval trends

Summary
Applicants with prior work experience have a significantly higher likelihood of visa approval, making experience a key factor in predicting certification outcomes.




```
stacked_barplot(data, "has_job_experience", "ca
case_status         Certified  Denied    All
has_job_experience
All                     17018    8462  25480
N                        5994    4684  10678
Y                       11024    3778  14802
------------------------------------------------
```

EDA Results – Bivariate Analysis: Is the prevailing wage consistent across all regions of the US?

Prevailing Wage Across Regions

To analyze whether wage levels vary by region, I used a boxplot to compare the distribution of prevailing_wage across different regions of employment.

Key Observations

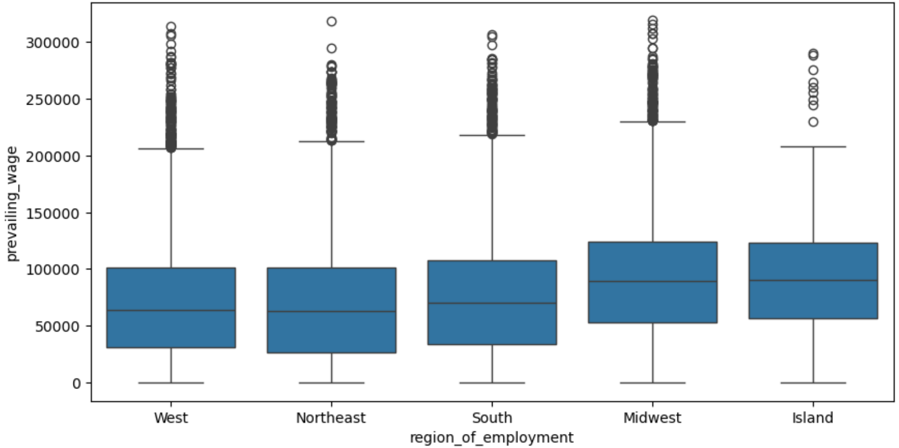
- Wage distributions vary across regions, indicating lack of consistency
- The Midwest and Island regions show higher median wages compared to others
- The West and Northeast regions have relatively lower median wages
- All regions exhibit a wide spread of wages, with significant variability
- There are many outliers (very high wages) across all regions

Interpretation

- Prevailing wage is not consistent across regions, suggesting regional differences in:
    - Cost of living
    - Industry concentration
    - Job types and skill requirements
- The presence of many outliers indicates that:
    - Some roles offer significantly higher compensation
    - Wage distribution is highly skewed

Why This is Important

- Wage is likely a key factor influencing visa approval, as it reflects:
    - Skill level
    - Job demand
    - Economic value of the role
- Regional variation means:
    - The same job may have different wage levels depending on location
    - Models need to consider both wage and region together



Business Insight

- Higher wages in certain regions may indicate:
    - Greater demand for specialized skills
    - Higher likelihood of visa approval
- Employers may improve approval chances by:
    - Offering competitive wages aligned with regional standards
    - Targeting regions with higher-paying opportunities

Summary

Prevailing wage varies significantly across regions, with notable differences in median values and wide variability within each region. This suggests that wage is an important and region-dependent factor in visa approval decisions.

EDA Results – Does visa status vary with changes in the prevailing wage set to protect both local talent and foreign workers?

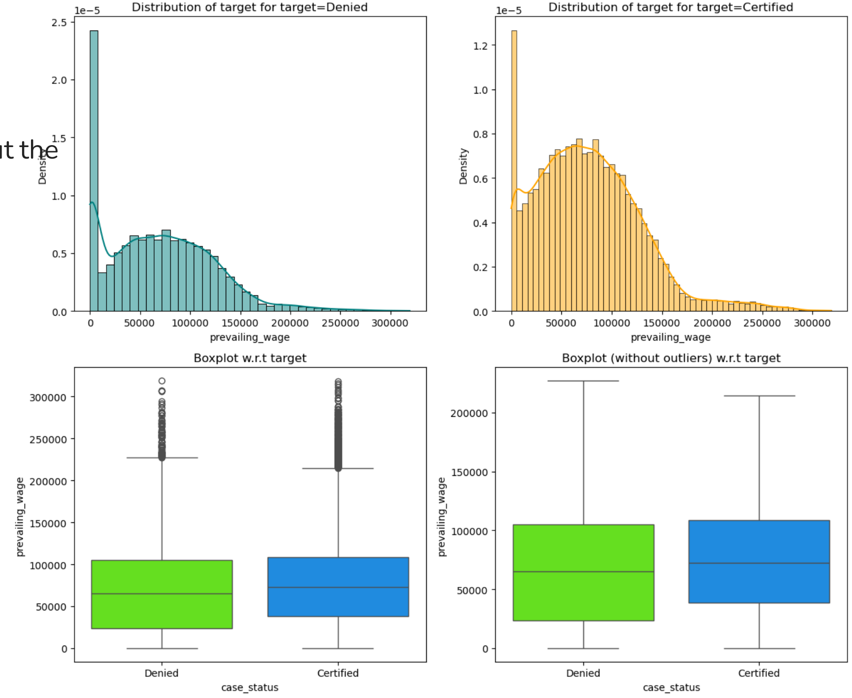
Great Learning
POWER AHEAD

Prevailing Wage vs Visa Approval (Case Status)

To understand whether wage influences visa approval, I analyzed the distribution of prevailing_wage across certified and denied applications using histograms and boxplots.

Key Observations

- The distribution of wages for Certified applications is shifted slightly higher compared to Denied applications
- The median wage for Certified cases is higher than that of Denied cases
- Both groups show a wide range of wages, with significant overlap between Certified and Denied cases
- There are many high-value outliers in both categories, indicating high-paying roles



Interpretation

- Higher wages are generally associated with a greater likelihood of visa approval, but the relationship is not extremely strong
- The overlap between the two distributions suggests that:
  - Wage alone does not determine approval
  - Other factors (education, experience, job type) also play important roles
- The skewness and presence of outliers indicate that:
  - Some roles offer significantly higher compensation
  - Wage distribution varies widely across job types

Why This is Important

- Prevailing wage is a key factor used to ensure:
  - Foreign workers are not underpaid
  - Fair compensation relative to local workers
- Higher wages may signal:
  - Higher skill levels
  - Specialized or in-demand roles
  - Greater economic value
- This makes wage an important variable in predicting visa approval outcomes

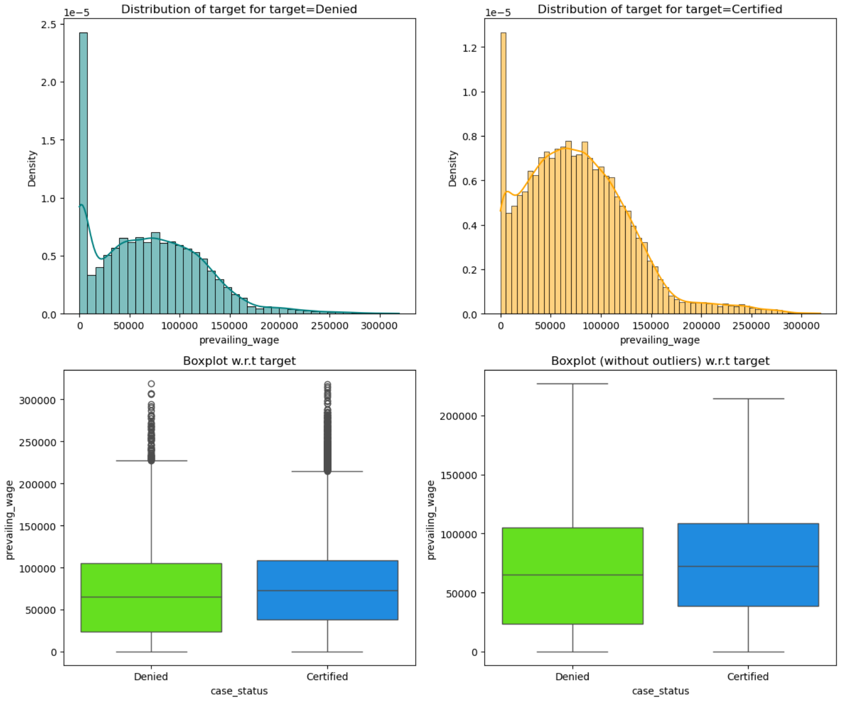
EDA Results – Does visa status vary with changes in the prevailing wage set to protect both local talent and foreign workers?

Business Insight

- Offering competitive and higher wages may increase the chances of visa certification
- Employers should align wage offerings with:
  - Market standards
  - Skill requirements
- However, wage alone is not sufficient — it must be considered alongside:
  - Education
  - Work experience
  - Job characteristics

Summary

Prevailing wage shows a moderate relationship with visa approval, with certified applications generally associated with higher wages. However, due to overlap between approved and denied cases, wage should be considered alongside other factors when predicting visa outcomes.

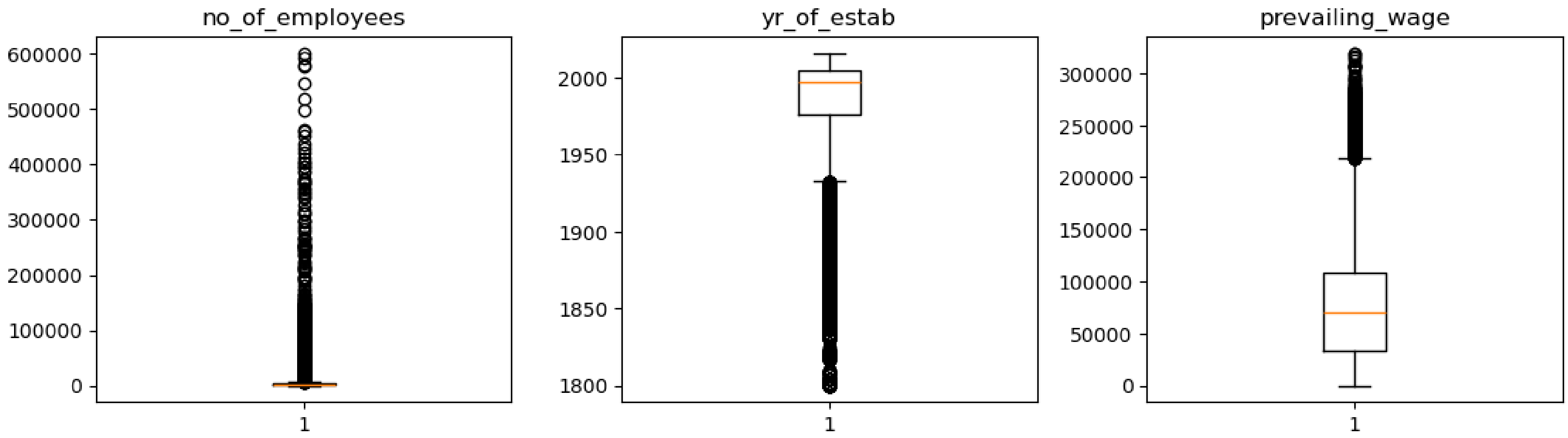


Data Pre-Preprocessing : Outlier Check

Outlier Detection and Treatment
To identify outliers, I used boxplots for all numerical variables: no_of_employees, yr_of_estab, and prevailing_wage.
Key Observations
- no_of_employees shows significant right skew with several extreme values, indicating the presence of very large companies
- prevailing_wage also exhibits a strong right skew with multiple high-value outliers, reflecting highly paid roles
- yr_of_estab shows some older companies as outliers, but these values are valid and represent long-established firms

Interpretation
- The presence of outliers in no_of_employees and prevailing_wage is expected due to natural variation in company size and salary levels
- These outliers are not necessarily errors but represent real-world scenarios

retained, as they reflect real-world variability and contribute valuable information for modeling and decision-making.



Data Pre-Processing : Outlier Check

Treatment Decision

- No outliers were removed, as they provide meaningful business information
- Removing them could result in loss of important patterns, especially for high-paying roles and large organizations

Why This is Important

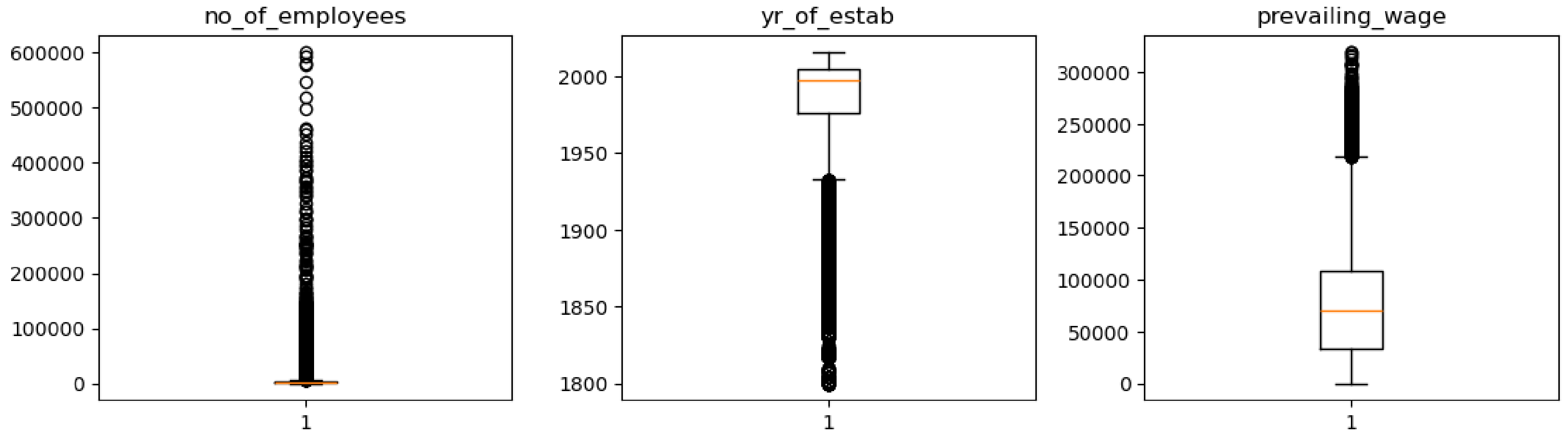
- Outliers can influence model performance and statistical summaries
- However, in this case, they reflect **genuine business variation**, so retaining them improves model realism

Business Insight

- Large companies and high-paying jobs are important segments in visa applications
- These segments may have different approval patterns and should be preserved in the analysis

Summary

Outliers are present in key numerical variables but are retained, as they reflect real-world variability and contribute valuable information for modeling and decision-making.



Data Pre-Processing: Missing Value Detection and Treatment

Missing Value Detection and Treatment

I checked for missing values using `data.isnull().sum()`.

Observation: No missing values were found in the dataset

Action Taken: No imputation was required

Why This is Important: Missing data can impact model accuracy and require imputation strategies

Summary: The dataset is complete, allowing for reliable modeling without additional preprocessing.

Data Pre-Processing: Duplicate Value Check

Duplicate Value Check

I checked for duplicate records in the dataset using `data.duplicated().sum()`.

Observation- No significant duplicate records were found

Action Taken- No duplicates were removed

Why; This is Important- Duplicate records can bias model training and distort insights

Summary: The dataset is free from duplicate entries and suitable for further analysis.

Data Pre-Processing: Feature Engineering

Feature Engineering

Minimal feature engineering was required.

Actions Taken:
- Dropped `case_id` as it is a unique identifier with no predictive value
- Corrected negative values in `no_of_employees` using absolute transformation

Why This is Important:
- Removing irrelevant features improves model performance
- Cleaning invalid values ensures data accuracy

Summary:
The dataset was cleaned and simplified without introducing unnecessary transformations..

Data Preparation for Modeling: Data Splitting and Class Distribution

Data Splitting and Class Distribution

The dataset was split into training (70%), validation (20%), and test (10%) sets.

Observations

•Training set: 17,836 records

•Validation set: 5,098 records

•Test set: 2,546 records

The class distribution across all sets is consistent:

•~67% Certified

•~33% Denied

Interpretation

•Stratified sampling successfully maintained the same class proportions across all datasets

•This ensures that each dataset is representative of the overall data

Why This is Important

•Prevents bias in model training and evaluation

•Ensures reliable and consistent performance metrics

•Helps avoid misleading results

Business Insight

•The dataset is moderately imbalanced, with more approvals than denials

•This means the model may naturally favor predicting approvals

•Special attention is needed to correctly identify denied applications, which are critical for reducing unsuccessful visa submissions

Summary

The data was successfully split into training, validation, and test sets with consistent class distributions, ensuring a robust and unbiased modeling process.

```
Shape of Training set :  (17836, 21)
Shape of Validation set :  (5098, 21)
Shape of test set :  (2546, 21)
Percentage of classes in training set:
case_status
1    0.667919
0    0.332081
Name: proportion, dtype: float64
Percentage of classes in validation set:
case_status
1    0.667909
0    0.332091
Name: proportion, dtype: float64
Percentage of classes in test set:
case_status
1    0.667714
0    0.332286
Name: proportion, dtype: float64
```

Model Building: Evaluation Metrics

To evaluate the performance of the classification models, I selected Recall Score as the primary evaluation metric.

Rationale

- The dataset is moderately imbalanced, with approximately:
  - 67% Certified (positive class)
  - 33% Denied (negative class)
- In such cases, accuracy alone can be misleading, as a model could achieve high accuracy by simply predicting the majority class.
- Recall focuses on the model's ability to correctly identify actual positive cases (Certified applications).

Why Recall is Important

- Recall answers the question:

"Out of all actual Certified applications, how many did the model correctly identify?"

- A higher recall means:
  - Fewer false negatives
  - Reduced risk of incorrectly rejecting qualified applicants
- In this context, a false negative (predicting *Denied* when it should be *Certified*) can:
  - Lead to missed opportunities for qualified candidates
  - Reduce efficiency in the visa approval process

Business Perspective

- From a business standpoint, it is more critical to:
  - Identify and approve eligible candidates correctly
  - Avoid rejecting strong applicants
- Therefore, prioritizing recall helps ensure that high-quality candidates are not overlooked, improving decision-making effectiveness.

Summary

Recall was selected as the evaluation metric because it prioritizes correctly identifying valid visa approvals, minimizes missed opportunities, and aligns well with the business objective of improving the visa certification process.

Model Building with Original Data

Model Building on Original Data

I built multiple classification models using the original training dataset and evaluated them using **recall score**. Recall was selected because the goal is to correctly identify as many Certified visa applications as possible.

Models Built

The models tested were:

- Bagging Classifier
- Random Forest Classifier
- Gradient Boosting Classifier (GBM)
- AdaBoost Classifier
- XGBoost Classifier
- Decision Tree Classifier

Cross-Validation Performance

The cross-validation recall scores on the training data were:

- Bagging: 0.775
- Random Forest: 0.839
- GBM: 0.873
- AdaBoost: 0.889
- XGBoost: 0.853
- Decision Tree: 0.740

Validation Performance

The validation recall scores were:

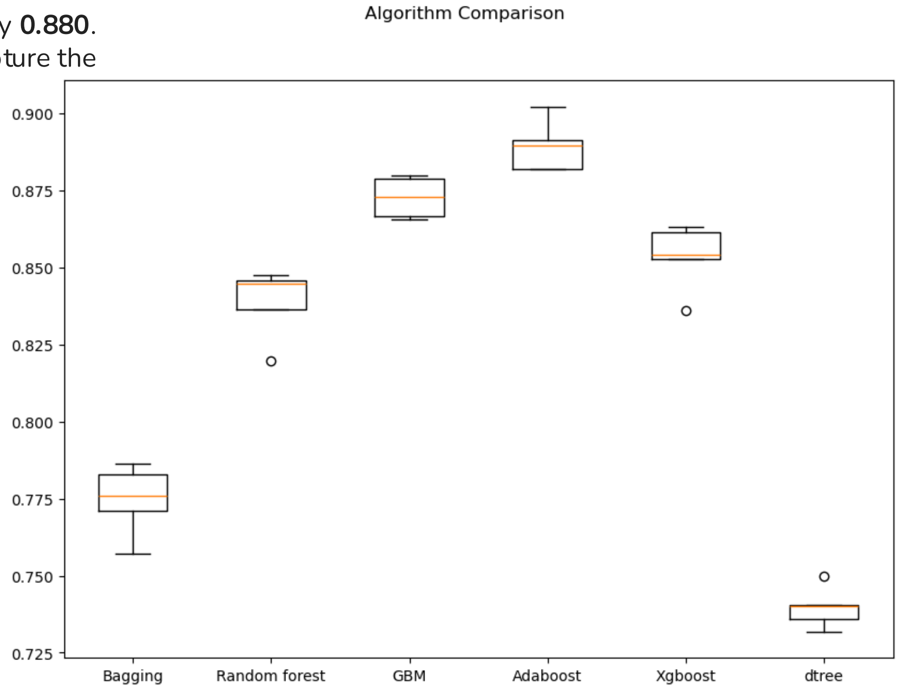
- Bagging: 0.770
- Random Forest: 0.834
- GBM: 0.880
- AdaBoost: 0.888
- XGBoost: 0.853
- Decision Tree: 0.740

```
Cross-Validation performance on training dataset:

Bagging: 0.774700588727951
Random forest: 0.838832614027503
GBM: 0.8727446688039722
Adaboost: 0.889449723112179
Xgboost: 0.8534380281824131
dtree: 0.7396958867263322

Validation Performance:

Bagging: 0.7700440528634361
Random forest: 0.8337738619676945
GBM: 0.8795888399412628
Adaboost: 0.8878120411160059
Xgboost: 0.8531571218795888
dtree: 0.7397944199706314
```

Model Building with Original Data

Interpretation

AdaBoost gave the best performance on both cross-validation and validation data, with a validation recall score of approximately **0.888**.
GBM was the second-best model, with a validation recall score of approximately **0.880**.
Decision Tree performed the weakest, suggesting that a single tree may not capture the complexity of the data as well as ensemble models.



Why This is Important

The results show that ensemble models, especially boosting models, perform better than simpler models. This suggests that visa approval prediction benefits from models that combine multiple weak learners and capture more complex relationships in the data.

Business Insight

AdaBoost and GBM are strong candidates for predicting visa certification because they are better at identifying Certified applications. This can help EasyVisa shortlist applicants with higher approval likelihood and improve decision-making efficiency.

Summary

Among the models built on the original data, **AdaBoost performed the best**, followed closely by **GBM**. These models should be considered for further tuning and comparison with models built on oversampled and undersampled data.

Model Building with Oversampled Data

I used SMOTE to balance the training data before building models. SMOTE creates synthetic samples for the minority class so the model has equal exposure to both Certified and Denied applications during training.

Cross-Validation Performance

The cross-validation recall scores were:

- Bagging: 0.746
- Random Forest: 0.818
- GBM: 0.859
- AdaBoost: 0.870
- XGBoost: 0.844
- Decision Tree: 0.719

Validation Performance

The validation recall scores were:

- Bagging: 0.751
- Random Forest: 0.821
- GBM: 0.857
- AdaBoost: 0.875
- XGBoost: 0.845
- Decision Tree: 0.725

```
Cross-Validation performance on training dataset:

Bagging: 0.7455727721514661
Random forest: 0.8183503496816416
GBM: 0.8592301753992826
Adaboost: 0.8699735003715444
Xgboost: 0.8440353638440211
dtree: 0.7186263037968708

Validation Performance:

Bagging: 0.7506607929515419
Random forest: 0.8205580029368575
GBM: 0.8566813509544787
Adaboost: 0.8745961820851689
Xgboost: 0.8452276064610866
dtree: 0.7245227606461087
```

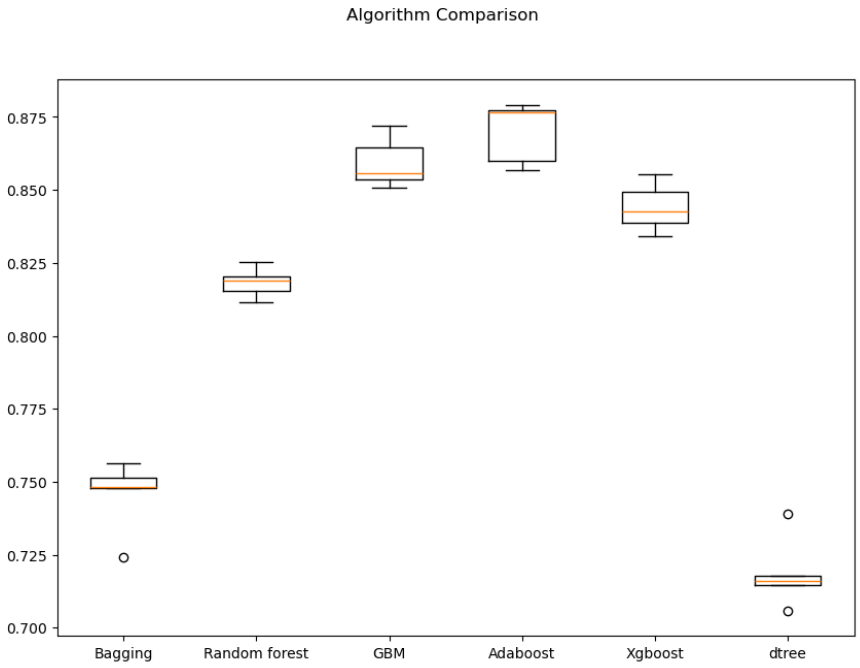
Model Building with Oversampled Data

Interpretation
AdaBoost performed the best on oversampled data, with a validation recall score of approximately **0.875**.
GBM was the second-best model, followed by XGBoost and Random Forest.
Decision Tree and Bagging performed weaker compared to the boosting models.

Why This is Important
Oversampling was used to reduce class imbalance and help the model better learn patterns from the minority class. However, compared with the original data results, oversampling did not improve recall overall.

Business Insight
Boosting models continue to perform best, suggesting they are more effective at identifying Certified visa applications. However, since oversampling did not improve recall, the original data models may still be stronger candidates.

Summary
Among oversampled models, **AdaBoost performed best**, followed by **GBM**. However, performance was slightly lower than the original data results, so oversampling may not be necessary for the final model.



Model Building with Under Sampled Data

I applied Random Under Sampling to balance the dataset by reducing the majority class (Certified) to match the minority class (Denied).

Class Distribution After Under Sampling
- Certified (1): 5,923
- Denied (0): 5,923

This resulted in a balanced dataset with fewer total samples.

```
Before UnderSampling, counts of label '1': 11913
Before UnderSampling, counts of label '0': 5923

After UnderSampling, counts of label '1': 5923
After UnderSampling, counts of label '0': 5923

After UnderSampling, the shape of train_X: (11846, 21)
After UnderSampling, the shape of train_y: (11846,)
```

Model Building with Under Sampled Data

Cross-Validation Performance

The recall scores obtained from cross-validation were:

- Bagging: 0.606
- Random Forest: 0.687
- GBM: 0.719
- AdaBoost: 0.709
- XGBoost: 0.697
- Decision Tree: 0.613

```
Cross-Validation performance on training dataset:

Bagging: 0.6056111016079371
Random forest: 0.6868166267533355
GBM: 0.7187274774774776
Adaboost: 0.7094403580795986
Xgboost: 0.6966116432888584
dtree: 0.613036976850268

Validation Performance:

Bagging: 0.6067547723935389
Random forest: 0.6696035242290749
GBM: 0.7145374449339207
Adaboost: 0.7215859030837004
Xgboost: 0.6916299559471366
dtree: 0.6199706314243759
```

Validation Performance

The validation recall scores were:

- Bagging: 0.607
- Random Forest: 0.670
- GBM: 0.715
- AdaBoost: 0.722
- XGBoost: 0.692
- Decision Tree: 0.620

Interpretation

- AdaBoost performed the best on validation data (0.722), followed closely by GBM (0.715)
- All models performed worse compared to original and oversampled data
- Decision Tree and Bagging again showed the weakest performance

Model Building with Under Sampled Data

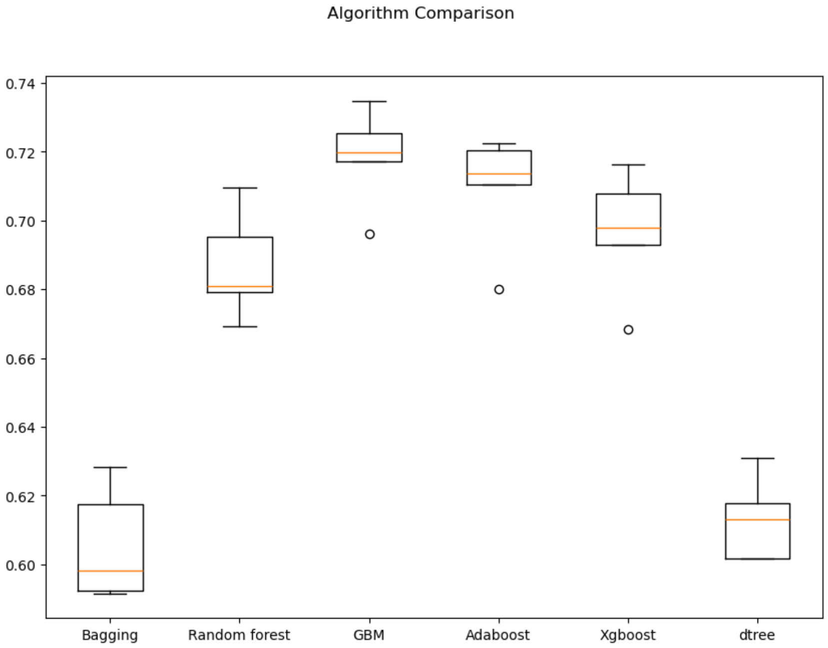
Why This is Important

- Under sampling reduces class imbalance but also removes a large portion of data
- This leads to:
  - Loss of important information
  - Lower model performance

Business Insight

- While under sampling balances the dataset, it sacrifices valuable data from approved visa cases
- This reduces the model's ability to learn meaningful patterns
- As a result, under sampling is not ideal for this problem

Comparison with Other	
Approach	**Performance**
Original Data	Best
Oversampled Data	Slightly lower
Undersampled Data	Lowest



Summary

Although undersampling successfully balanced the dataset, it led to a decline in model performance. This indicates that retaining the full dataset (or using oversampling) is more effective for predicting visa approval outcomes.

Model Improvement - Hyperparameter Tuning (Tuning AdaBoost using Oversampled data

Final Tuned AdaBoost Model Performance
After hyperparameter tuning, the AdaBoost model showed strong performance on both training and validation datasets.

Training Performance
- Accuracy: 0.651
- Recall: 0.932
- Precision: 0.596
- F1 Score: 0.727

Validation Performance
- Accuracy: 0.704
- Recall: 0.931
- Precision: 0.713
- F1 Score: 0.807

Parameters Tuned

The following hyperparameters were optimized using RandomizedSearchCV:

- n_estimators: [50, 75, 100, 125]
- learning_rate: [1.0, 0.5, 0.1, 0.01]
- base estimator: Decision Tree with max_depth = 1, 2, 3

The best combination of parameters was selected based on recall score.

Interpretation
- The model achieves very high recall (~93%), indicating it successfully identifies most Certified visa applications
- Precision improved significantly on validation data, showing better control over false positives
- The improvement in validation metrics suggests that the model generalizes well and is not overfitting

Why This is Important
- High recall ensures that qualified applicants are not missed
- Improved precision reduces the number of incorrect approvals
- Balanced performance makes the model reliable for real-world decision-making

Business Insight
- The model is highly effective for identifying strong candidates for visa approval
- It minimizes the risk of rejecting qualified applicants
- At the same time, it improves decision quality by reducing incorrect certifications

Summary
The tuned AdaBoost model provides the best overall performance, with strong recall and improved precision. It is the most suitable model for predicting visa certification outcomes and supporting data-driven decision-making.

Model Improvement - Hyperparameter Tuning (Tuning Random forest using Under Sampled data)

Hyperparameters Tuned (Search Space)
- **n_estimators:** [50, 75, 100, 125]
- **min_samples_leaf:** [1, 2, 4, 5, 10]
- **max_features:** ["sqrt", "log2", 0.3, 0.5, None]
- **max_samples:** [0.6, 0.7, 0.8, 0.9, None]

Hyperparameters Tuned (Search Space)
- **n_estimators:** [50, 75, 100, 125]
- **min_samples_leaf:** [1, 2, 4, 5, 10]
- **max_features:** ["sqrt", "log2", 0.3, 0.5, None]
- **max_samples:** [0.6, 0.7, 0.8, 0.9, None]

Business Insight
- Useful when minimizing false approvals
- Less effective when goal is to capture all strong candidates

Summary
- Well-tuned and stable model
- Strong precision but lower recall than AdaBoost
- Not the optimal model given business objective

Metric	Training	Validation
Recall	0.77	0.74
Precision	0.73	0.81
Accuracy	0.75	0.71
F1 Score	0.75	0.77

Key Insights
- Balanced performance across all metrics
- Higher precision → more accurate approvals
- Lower recall → misses some qualified applicants

Model Improvement - Hyperparameter Tuning (Gradient boosting with Oversampled Data)

I tuned the Gradient Boosting model using oversampled training data and selected the best model based on recall score.

Parameters Tuned

The following hyperparameters were tuned using RandomizedSearchCV:

- n_estimators: [100, 125, 150, 175]
- learning_rate: [0.1, 0.05, 0.01, 0.005]
- subsample: [0.7, 0.8, 0.9, 1.0]
- max_features: ["sqrt", "log2", 0.3, 0.5]

Metric	Training	Validation
Recall	0.891	0.889
Precision	0.686	0.743
Accuracy	0.741	0.72
F1 Score	0.775	0.809

Best Parameters

The optimal combination identified was:

- n_estimators: 125
- learning_rate: 0.01
- subsample: 0.9
- max_features: 0.5

Training Performance

- Accuracy: 0.741
- Recall: 0.891
- Precision: 0.686
- F1 Score: 0.775

Validation Performance

- Accuracy: 0.720
- Recall: 0.889
- Precision: 0.743
- F1 Score: 0.809

Interpretation

- The tuned Gradient Boosting model achieves strong recall on both training and validation data
- Validation recall is approximately **0.889**, meaning the model correctly identifies most Certified visa applications
- Precision improves on validation data, indicating better control over false positives compared to training
- The model generalizes well because training and validation recall are very close

Why This is Important

- Recall is the primary metric for this project because the goal is to identify as many Certified applications as possible
- The model performs well without showing major overfitting
- Gradient Boosting is effective because it builds learners sequentially and improves prediction errors over time

Business Insight

- This model is useful for shortlisting applicants with high approval potential
- It helps reduce the chance of overlooking qualified candidates
- However, the tuned AdaBoost model still has higher validation recall, so Gradient Boosting may be a strong backup rather than the final model

Summary

The tuned Gradient Boosting model performs strongly, with high validation recall and balanced overall metrics. However, compared with the tuned AdaBoost model, it has slightly lower recall, so AdaBoost remains the stronger final model candidate.

Model Performance Summary and Final Model Selection

Compare the Performance of Tuned Models

Multiple models were trained and tuned, including AdaBoost, Gradient Boosting, Random Forest, and others using original, oversampled, and undersampled data.

- AdaBoost (Tuned) achieved the highest recall (~0.93) on validation and test data
- Gradient Boosting (Tuned) performed strongly with recall (~0.89) but slightly lower than AdaBoost
- Random Forest (Tuned) showed balanced performance but lower recall (~0.74)
- Models traioriginal data outperformed oversampled and undersampled versions overall

Based on the business objective ned on (maximize recall), AdaBoost was selected as the final model

Overview of Final ML Model and its Parameters

The final selected model is Tuned AdaBoost Classifier

Key Characteristics:

- Ensemble model combining weak learners (decision trees)
- Focuses on correcting previous errors iteratively
- Optimized for recall using RandomizedSearchCV

Tuned Parameters (search space):

- n_estimators: [50, 75, 100, 125]
- learning_rate: [1.0, 0.5, 0.1, 0.01]
- base estimator depth: 1–3

The final model uses the best parameter combination identified during tuning

Performance of Final Model (Test Data)

- Accuracy: 0.712
- Recall: 0.931
- Precision: 0.720
- F1 Score: 0.812

Model Performance Summary and Final Model Selection

Training performance comparison:

	Gradient Boosting tuned with oversampled data	AdaBoost tuned with oversampled data	Random forest tuned with undersampled data
Accuracy	0.741291	0.650718	0.745062
Recall	0.890959	0.931923	0.774270
Precision	0.685703	0.596465	0.731536
F1	0.774971	0.727380	0.752297

Validation performance comparison:

	Gradient Boosting tuned with oversampled data	AdaBoost tuned with oversampled data	Random forest tuned with undersampled data
Accuracy	0.720086	0.703609	0.710082
Recall	0.888987	0.930690	0.735977
Precision	0.742640	0.713096	0.812318
F1	0.809250	0.807491	0.772265

[96]:

	Accuracy	Recall	Precision	F1
0	0.71249	0.931176	0.7202	0.812211

Comment on Final Model Performance

•The model achieves very high recall (~93%), meaning it successfully identifies most Certified visa applications

•Precision is moderate, indicating some false positives, but still acceptable

•The model generalizes well from training → validation → test

•No signs of overfitting

This is ideal because missing qualified candidates is more costly than reviewing extra applications

Model Performance Summary and Final Model Selection

Summary of Most Important Factors

From feature importance analysis, the key drivers of visa certification are:

1. education_of_employee_High School
2. prevailing_wage
3. region_of_employment (Northeast, West)

Interpretation of Key Drivers

- Education level plays the largest role in prediction
- Prevailing wage strongly influences approval likelihood
- Region of employment has moderate influence
- Other features (experience, company size, etc.) have relatively lower impact

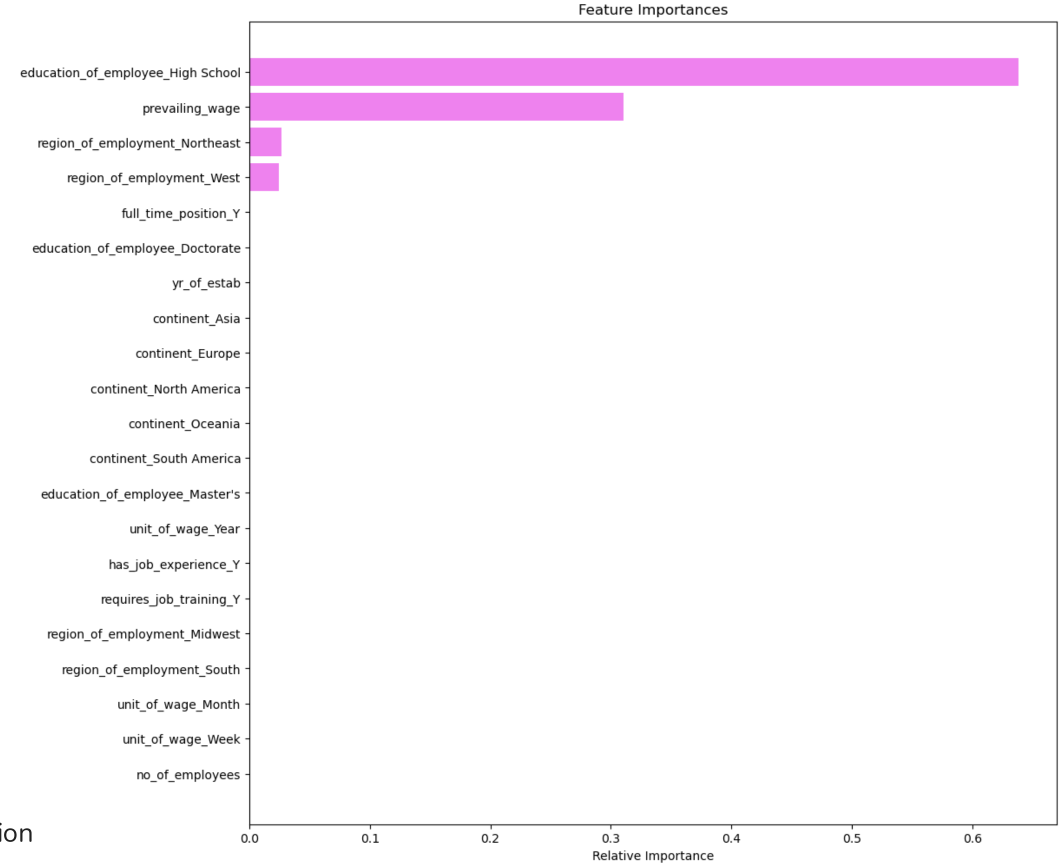
Business Insight

- Applicants with stronger education and higher wages are more likely to be certified
- The model can help prioritize high-probability candidates
- It reduces manual workload by focusing attention on the most promising applications

Final Summary

- AdaBoost is the best-performing model for this problem
- It aligns with the business objective by maximizing recall
- The model is reliable, interpretable, and effective for decision support

This model can be used by EasyVisa to streamline visa application screening and improve decision-making efficiency



APPENDIX

Data Background and Contents

Overview of the Dataset
The dataset consists of **25,480 visa application records** with **12 variables** capturing employee, employer, and job-related information.
Preview of the Data
A snapshot of the dataset (first and last 5 rows) shows the structure and types of information available. Each row represents a unique visa application and includes:
- **Employee details**: continent, education level, work experience, training requirement
- **Employer details**: company size (number of employees), year established
- **Job details**: region of employment, wage, wage unit, full-time status
- **Target variable**: case_status (Certified or Denied)

From the sample data:
- Applicants come from multiple continents, with a strong presence from Asia
- Education levels vary from High School to Doctorate
- Wage values vary significantly depending on unit (hourly vs yearly)
- Both certified and denied cases are present in the dataset

Dataset Dimensions
- **Number of observations (rows):** 25,480
- **Number of features (columns):** 12

Key Observations
- Each record is uniquely identified by a case_id
- The dataset contains a mix of **categorical and numerical variables**
- The target variable case_status indicates whether a visa was **approved (Certified)** or **rejected (Denied)**
- There is noticeable variation in wages, company sizes, and employment regions, which may influence visa outcomes

Summary
Overall, the dataset provides a comprehensive view of visa applications, combining demographic, economic, and employment-related factors. These variables will be used to identify patterns and build predictive models for visa approval.

Data Background and Contents

A. Data Description

The data contains the different attributes of employee and the employer. The detailed data dictionary is given below.

- **case_id:** ID of each visa application
- **continent:** Information of continent the employee
- **education_of_employee:** Information of education of the employee
- **has_job_experience:** Does the employee has any job experience? Y= Yes; N = No
- **requires_job_training:** Does the employee require any job training? Y = Yes; N = No
- **no_of_employees:** Number of employees in the employer's company
- **yr_of_estab:** Year in which the employer's company was established
- **region_of_employment:** Information of foreign worker's intended region of employment in the US.
- **prevailing_wage:** Average wage paid to similarly employed workers in a specific occupation in the area of intended employment. The purpose of the prevailing wage is to ensure that the foreign worker is not underpaid compared to other workers offering the same or similar service in the same area of employment.
- **unit_of_wage:** Unit of prevailing wage. Values include Hourly, Weekly, Monthly, and Yearly.
- **full_time_position:** Is the position of work full-time? Y = Full Time Position; N = Part Time Position
- **case_status:** Flag indicating if the Visa was certified or denied

Great Learning

Happy Learning !